Huy Le

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Education

University of Alberta — Edmonton, AB

Bachelor of Science in Computer Science, Minor in Mathematics — *Expected Dec. 2027*

- **Coursework:** Data Structures & Algorithms, Database Systems, Operating Systems, Software Engineering

Technical Skills

Languages: Java, Python, TypeScript, JavaScript, SQL, C, HTML/CSS
Frameworks: Spring Boot, Spring Data JPA, React, Next.js, FastAPI, Django, Flask, Node.js
Data & Messaging: PostgreSQL, Redis, MongoDB, RabbitMQ, Kafka, Firestore, NoSQL
AI / ML: Model Context Protocol (MCP), LangChain, RAG, GraphRAG, vector embeddings
Tools & Cloud: Docker, Maven, Flyway, Git/GitHub, GitHub Actions, AWS, GCP, Linux, pytest, CI/CD
Protocols: REST APIs, WebSockets, CAN, SAE J1939

Experience

Embedded Software Research Assistant — Jul. 2026 – Present

Southern Taiwan University of Science and Technology (TEEP) — *Tainan, Taiwan*

- Designed a CAN-based instruction protocol using SAE J1939 conventions to configure an STM32 pattern generator, and develop Python tooling to encode and transmit command frames through the TSMaster interface
- Built a signal-configuration UI that converts engineer-selected values into encoded command payloads, applying signal scaling, byte ordering, and range validation for **179 unique SPNs** to configure the device without modifying firmware

Applied AI Software Developer Intern — May 2025 – Aug. 2025

FPT Software — *Hanoi, Vietnam*

- Built and containerized a Model Context Protocol (MCP) server using Streamable HTTP for OpenAI tool integration, reducing end-to-end latency by ∼**35%** on an internal chatbot
- Optimized the LangChain and ChromaDB indexing pipeline with an embedding cache, eliminating redundant re-embedding and increasing indexing throughput by ∼**40%**
- Prototyped a knowledge-graph-augmented retrieval layer using GraphRAG, improving answer accuracy by ∼**18%** over the existing RAG baseline across a **100-question** evaluation set

Software Developer Intern — Jun. 2024 – Aug. 2024

CodeDream Education — *Hanoi, Vietnam*

- Replaced a manual spreadsheet-based scheduling workflow with a React and TypeScript admin dashboard backed by FastAPI, supporting operations for **50+ classes** across **4 education centres**
- Added pytest regression coverage, then profiled and refactored a legacy PHP and Python judging system, reducing grading latency by **20%** for **400+ students**
- Fine-tuned and deployed a CodeLlama classifier behind a Flask API to flag likely AI-generated submissions for instructor review, achieving **72% classification accuracy**

Software Developer Intern — May 2023 – Aug. 2023

DIBIZ – Digital Business — *Hanoi, Vietnam*

- Redesigned consumer batching and message-handling logic in a Python and RabbitMQ pipeline, reducing daily processing time by **15%** across data for **500,000 customers**

Projects

JobPulse | GitHub | Link | *Java/Spring Boot, PostgreSQL, RabbitMQ, Kafka, GCP* — Sep. 2026 – Present

- Built and deployed a **Java/Spring Boot job aggregator to GCP Compute Engine** using Docker and GitHub Actions; RabbitMQ workers processed **50,000 synthetic postings at 642 postings/second** in local benchmarks with no duplicates after replay.
- Implemented **PostgreSQL full-text search** with relevance ranking and filters, sustaining **100 requests/second at 24 ms p95 latency** across 50,000 postings in a five-minute local load test with zero request failures.
- Designed a **transactional outbox and Kafka consumers** for saved-search alerts, generating **10,000 alerts across 100 searches in under 3.6 seconds** in local burst tests; verified duplicate-safe replay of 100 events.

Fantasy Football | GitHub | *React, TypeScript, FastAPI, PostgreSQL, Redis, Docker* — Aug. 2026 – Present

- Built a full-stack Fantasy Premier League platform with React, TypeScript, FastAPI, and PostgreSQL, modeling deadline-locked squads, transfers, captaincy, chips, scoring, and automatic substitutions
- Designed an adaptive ingestion worker for official player, fixture, and live gameweek data and added Redis caching for player searches, reducing benchmarked latency from **15 ms to 6 ms** at a **60% cache-hit rate**